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UB-7449

Integrated M. C. A. (Sem. VI)
Examination, June - 2025

IMCA - 603T

ARTIFICIAL INTELLIGENCE

Time allowed : Three Hours

Max. Marks : 70

Min. Marks : 28

***Note : This question paper is divided into
two sections as under:***

SECTION - A

Max. Marks-20

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All questions will have equal marks. (10×2=20)

SECTION - B

Max. Marks-50

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Words limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks. (10×5=50)

SECTION - A

- 1 (i) Define 'Turing Test' with an example.
- (ii) Define 'Intelligent Agents'. Give its two functions.
- (iii) What do you understand by 'Problem characteristics'?
- (iv) Define 'Hill-climbing' search strategy, with an example.
- (v) Explain the 'Resolution principle'.
- (vi) Define 'frames problem'. Give its two types.

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2

[Contd...

- (vii) Define a 'Prolog program'. Give example.
- (viii) Explain 'Conditional probability'.
- (ix) Define 'Parsing'. Explain its different types.
- (x) Differentiate 'Context-free' and 'Transformational' Grammar with suitable examples.

SECTION - B

UNIT - I

- 2 (a) Define Artificial Intelligence. Give its application areas.
- (b) Explain the foundations of AI in detail.

OR

What are the four basic types of 'Agent Program' in any intelligent system? Explain how do you convert them into learning agents?

UNIT - II

- 3 Define A* search. Explain various stages of A* search with an example.

OR

Define constraint satisfaction problem. How constraint satisfaction problem is formulated as a search problem? Explain with an example.

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3

[Contd...

UNIT - III

- 4 ✓ Explain the various steps involved in knowledge engineering process. Give illustration with an example.

OR

- (a) List and explain the various drawbacks of propositional logic.
- (b) Explain the syntax and semantics in first order logic.

UNIT - IV

- 5 (a) Write a note on 'Truth-Maintenance systems.'
- (b) Describe the various applications of inference in AI.

OR

- (a) Define Monotonic Reasoning. Give its utility in AI.
- (b) What is a Bayesian network? Describe its various properties.

UNIT - V

- 6 ✓ What is NLP? Explain its advantages and roles of AI.

OR

Give the various applications of recursive transition networks in NLP.